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Eurobioc2026 BiocContainer Report

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Introduction

Containers enhance reproducibility and additionally offer a convenient way to share the setup used and needed to run a script. Offering a way to create containers which contain desired `R` packages from within `R` allows users to more easily create these containers and work with them. `R` packages like dockerfiler Fay et al. (2026) offer functions to create and work with Docker files within `R`. However, currently there is no convenient way to create Docker files and containers from `R` while also controlling for the bioconductor release. The aim in this hackathon was to create a minimal set of functions that offer this possibility, and this work eventually gave rise to the R/Bioconductor package `SimpleBiocContainer`.

R package `SimpleBiocContainer`

As part of the European Bioconductor Hackathon 2026, we created an R package called `SimpleBiocContainer` which offers a more user friendly way to create, build and push Docker containers to Dockerhub from R. A working version of the package can be found on GitHub: (https://github.com/lgatto/SimpleBiocContainer.git).

The functions in `SimpleBiocContainer` make use of the existing Bioconductor docker containers (see this link for more details), which already contain system dependencies and a few basic packages. On top of that, we add the user-specified Bioconductor/R packages using `BiocManager::install()`. The Bioconductor version, and thus the corresonding docker container, is automatically determined from the R session.

Currently, the package has two exported functions:

- makeSimpleBiocContainer creates a container directory and populates it with a Dockerfile and optional data and script directories.
- buildPushDocker builds and pushes the container to Dockerhub or optionally Github container registry directly from `R`.

Here is a simple example to illustrate its use:

```
library(SimpleBiocContainer)

container_path <- makeSimpleBiocContainer(
  package = "MsCoreUtils",
  container = "my_container",
  data = "data.rda",
  script = "script.R")
```

```
buildPushDocker(
  container = container_path,
  dockerUsername = "docker_username")
```

Conclusions / Future Work

The current version of the package covers all features that we wanted to include, but represents a simple first implementation that may require further refinement in the future.

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References

Fay, C., Guyader, V., Parry, J., & Rochette, S. (2026). *Dockerfiler: Easy dockerfile creation from r*. https://doi.org/10.32614/CRAN.package.dockerfiler

Author contributions